

Problem 2.8

Define $\mathcal{C}$ to be the smallest class closed under countable intersections/unions containing $\mathcal{A}$. Further we define $\mathcal{D} = \{A \mid A^c \in \mathcal{C}\}$. Then by property of $\mathcal{A}$ one has $\mathcal{A} \subset \mathcal{D}$ and further $\mathcal{D}$ is closed under countable intersections/unions: Let $A_1^c, A_2^c, \dots$ be in $\mathcal{C}$ (equivalently $A_1, A_2, \dots$ in $\mathcal{D}$), then one has

$$\left(\bigcup_n A_n\right)^c = \bigcap_n A_n^c \in \mathcal{C},$$

since $\mathcal{C}$ is closed under countable intersections and also

$$\left(\bigcap_n A_n\right)^c = \bigcup_n A_n^c \in \mathcal{C},$$

since $\mathcal{C}$ is closed under countable unions. Therefore

$$\bigcup_n A_n, \bigcap_n A_n \in \mathcal{D}$$

which is equivalent to $\mathcal{C}$ being closed under intersections. Therefore one has $\mathcal{A} \subset \mathcal{C} \subset \sigma(\mathcal{A})$ and $\mathcal{C}$ is closed under intersections/unions/complements, so therefore $\mathcal{C} = \sigma(\mathcal{A})$.